ECE 3355: Electronics

Exam 3

Complete the exam on your own, without the help of your notes, prior examples or solutions, your book, or any communication/interaction with others. You must write a complete solution that shows the relevant steps if you want full credit for the problem. You may use a calculator and a crib sheet is provided as part of this exam. **You will have 1 hr. 15 min. to finish the exam.**

Student's Name: ______________________________

Question	Points	Score
1	45	
2	55	
Total:	100	

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha}\right) e^{v_{BE}/V_T}$$

Note: For the pnp transistor, replace v_{BE} with v_{EB}.

$$i_C = \alpha i_E$$

$$i_C = \beta i_B$$

$$\beta = \frac{\alpha}{1-\alpha}$$

$$i_B = (1-\alpha)\, i_E = \frac{i_E}{\beta+1}$$

$$i_E = (\beta+1)\, i_B$$

$$\alpha = \frac{\beta}{\beta+1}$$

$V_T = \text{thermal voltage} = \dfrac{kT}{q} \simeq 25\,\text{mV}$ at room temperature

Summary of Table 7.3 (Small Signal Model Parameters)

Model Parameters in Terms of DC Bias Currents

$g_m = \dfrac{I_C}{V_T} \quad r_e = \dfrac{V_T}{I_E} = \alpha\dfrac{V_T}{I_C} \quad r_\pi = \dfrac{V_T}{I_B} = \beta\dfrac{V_T}{I_C} \quad r_o = \dfrac{|V_A|}{I_C}$

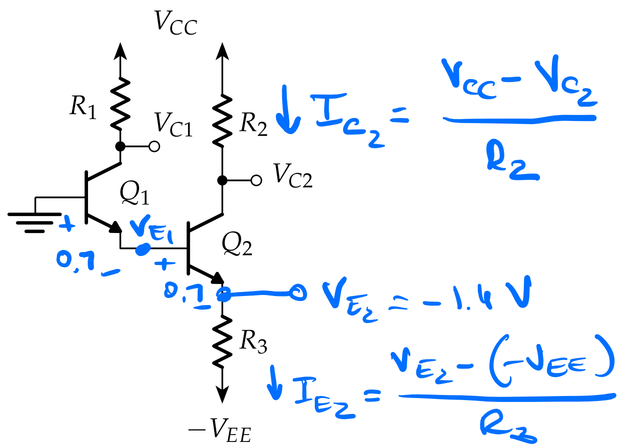
In Terms of g_m

$r_e = \dfrac{\alpha}{g_m} \qquad r_\pi = \dfrac{\beta}{g_m}$

In Terms of r_e

$g_m = \dfrac{\alpha}{r_e} \qquad r_\pi = (\beta+1)\, r_e \qquad g_m + \dfrac{1}{r_\pi} = \dfrac{1}{r_e}$

1. Consider the following circuit where $R_1 = 10\ \text{k}\Omega$, $R_3 = 300\ \Omega$, $V_{CC} = 14$ V, $-V_{EE} = -14$ V and β=99.



(a) (15 points) Find R_2 so that $V_{C2} = 1$ V. Is Q2 in an 'active' state?

$$I_{E_2} = \frac{-1.4 + 14}{300} = 42\text{mA}, \quad I_{C_2} = \frac{\beta}{\beta+1} \cdot I_{E_2} = \frac{99}{100} \cdot 42\text{mA} = 41.58\text{mA}$$

$$R_2 = \frac{V_{CC} - V_{C_2}}{I_{C_2}} = \frac{14-1}{41.58\text{m}} = 312.65\ \Omega$$

$$V_{C_2} - V_{E_2} = 1 - (-1.4) = 2.4 > 0.2 \rightarrow \underline{\text{active}}$$

(b) (15 points) Find V_{C1}. Is Q1 in an 'active' state?

$$I_{E_1} = I_{B_1} = \frac{I_{E_2}}{\beta+1} = \frac{42\text{mA}}{100} = 0.42\text{mA}$$

$$I_{C_1} = \frac{\beta}{\beta+1} I_{E_1} = \frac{99}{100} \cdot 0.42\text{mA} = 0.4158\text{mA}$$

$$V_{C_1} = V_{CC} - I_{C_1} \cdot R_1 = 14 - 0.4158\text{m} \cdot 10\text{k}$$

$$= 14 - 4.158 = 9.842\ \text{V}$$

$$V_{C_1} - V_{E_1} = 9.842 - (-0.7) > 0.2\text{V} \rightarrow \underline{\text{active}}$$

(c) (15 points) What values of R_2 ensure that Q2 is in an 'active' state?

$$V_{C_2} - V_{E_2} > 0.2\ V$$

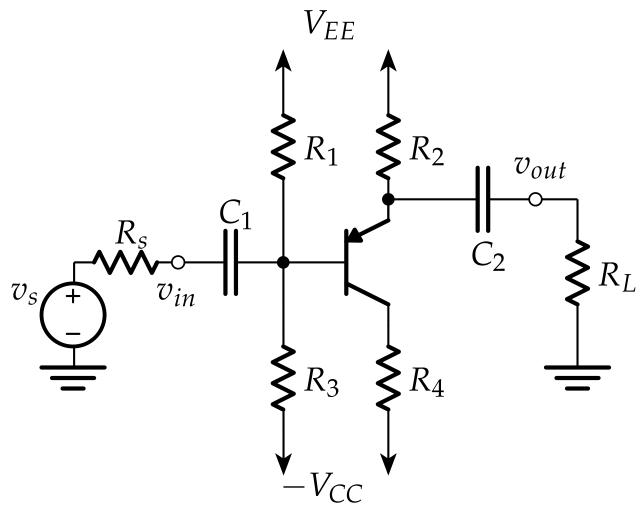
$$V_{CC} - I_{C_2} \cdot R_2 - V_{E_2} > 0.2$$

$$V_{CC} - 0.2 - V_{E_2} > I_{C_2} R_2$$

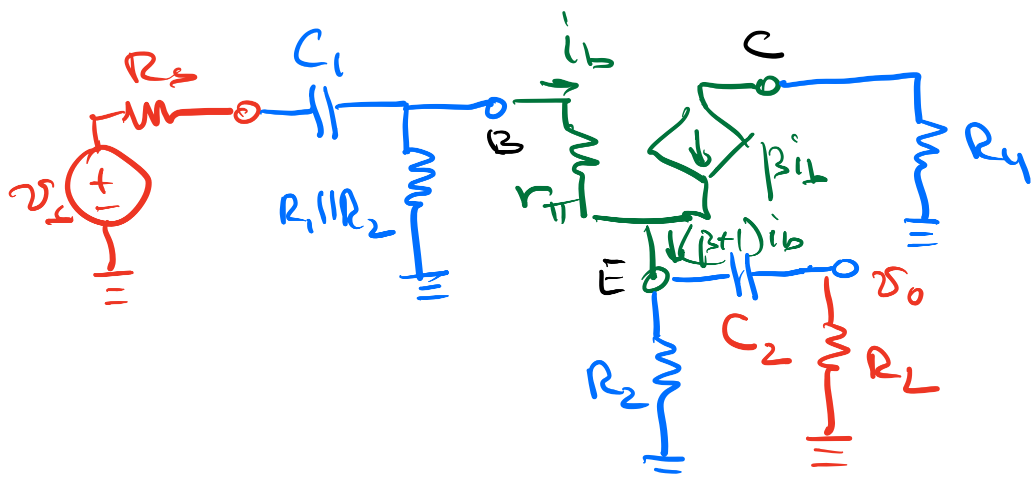
$$R_2 < \frac{V_{CC} - 0.2 - V_{E_2}}{I_{C_2}} = \frac{14 - 0.2 - (-1.4)}{41.58m}$$

$$R_2 < 365.6\ \Omega$$

3. Use the following circuit to complete this problem.

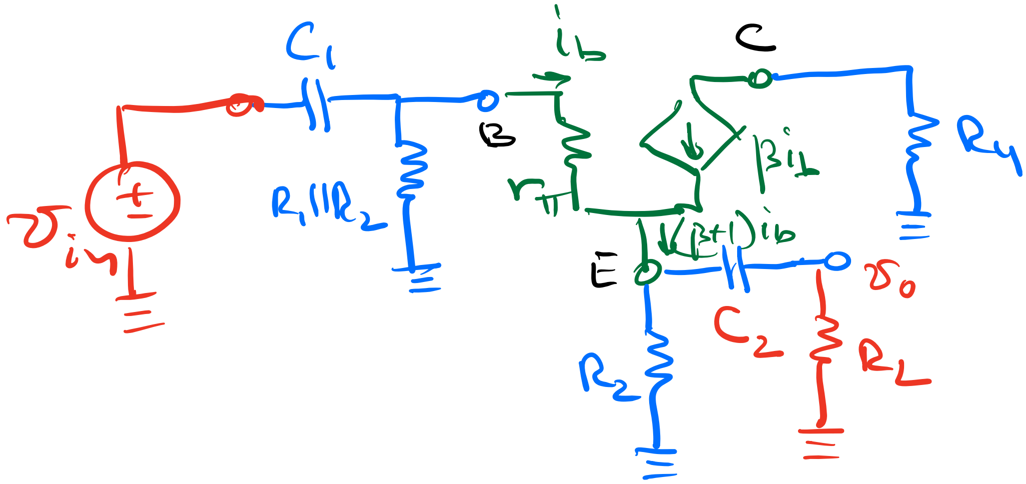


(a) (5 points) Assuming that the transistor is in an active mode, sketch the small signal model for the circuit. *Hint:* You do not need to complete a DC analysis - just the AC analysis.



(b) (10 points) Find the expression for the *midband gain*, A, with the load and source removed. The solution should be expressed in terms of resistors, β, r_π.

$$A = \frac{v_o}{v_{in}} = \frac{R_2(\beta+1)i_b}{r_\pi i_b + (\beta+1) i_b R_2} = \frac{R_2(\beta+1)}{r_\pi + (\beta+1)R_2}$$

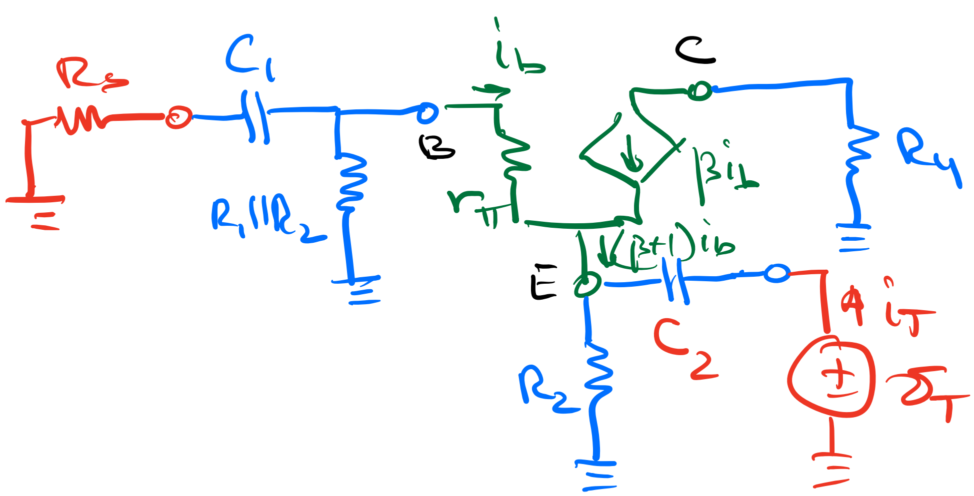


(c) (10 points) Find expression for the input resistance, R_{in}, for the *midband*.
Hint: attach the load.

$$R_{in} = \frac{v_{in}}{i_{in}} = R_1 || R_2 || \frac{r_\pi i_b + (\beta+1) i_b (R_2 || R_L)}{i_b}$$

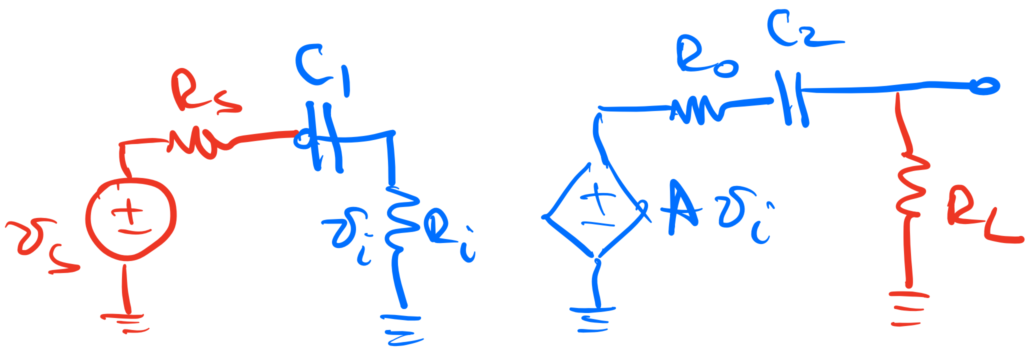
$$= R_1 || R_2 || \left(r_\pi + (\beta+1)(R_2 || R_L) \right)$$

(d) (10 points) Find expression for the output resistance, R_{out}, for the *midband*.
Hint: attach the source.



$$R_o = R_2 \| \frac{-r_\pi i_b - i_b (R_1 \| R_2 \| R_S)}{-(\beta+1) i_b} = R_2 \| \frac{r_\pi + R_1 \| R_2 \| R_S}{\beta + 1}$$

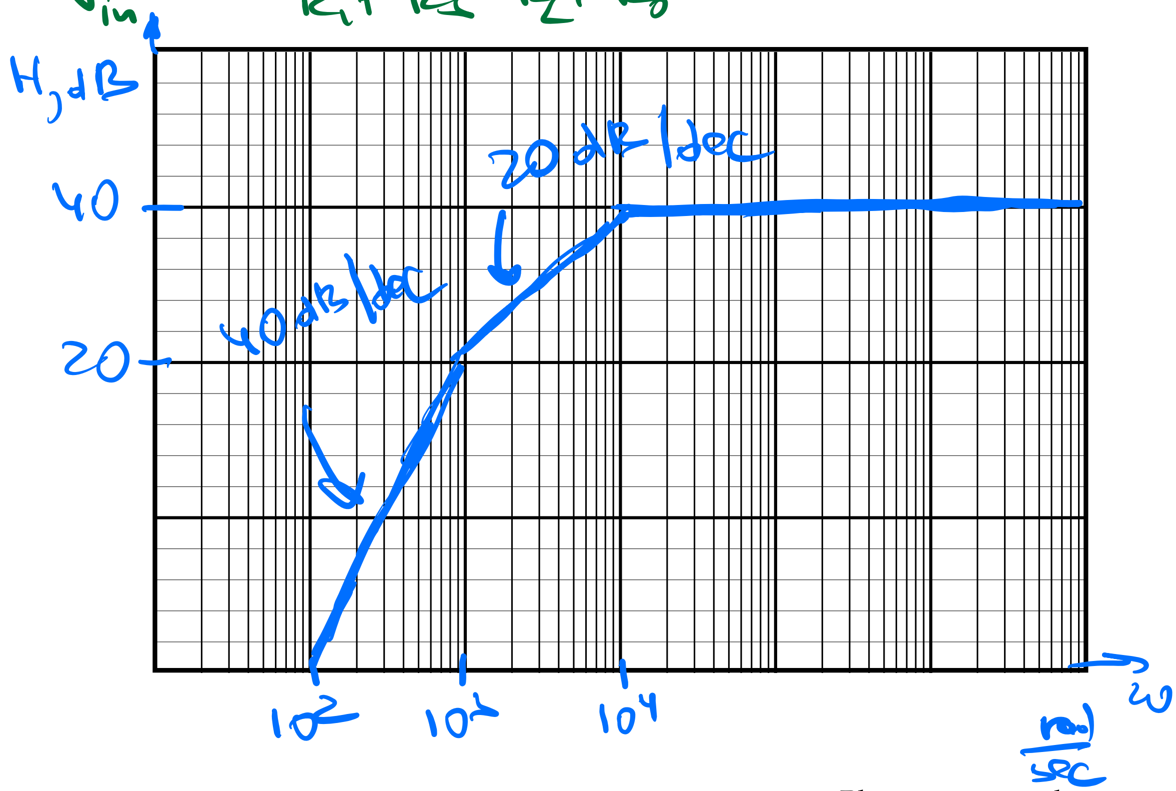
(e) (10 points) Draw the equivalent circuit of the amplifier using the parameters identified above (R_{in}, R_{out}, and A) with the source and load attached. Find expressions for resistances R_{C1} and R_{C2} as seen by capacitors C_1 and C_2, respectively.



$$R_{C_1} = R_S + R_i \qquad R_{C_2} = R_o + R_L$$

(f) (10 points) Assuming R_{in} = 5k, R_{out} = 5k, R_S = 5k, R_L= 5k, C_1 = 10nF, C_2 = 0.1uF, and midband gain A=400, plot the Bode plot of the transfer function for the circuit. Make sure to label the axes.

$$\frac{V_o}{V_{in}} = A \frac{R_i}{R_i + R_S} \frac{R_L}{R_L + R_o} = 100$$



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$P_1 = 10^4$

$P_2 = 10^3$

End of Exam